

1. Answer the questions about the function $f(x)$ graphed below.

(a) $\int_2^4 f(x) dx = A_{up} = \boxed{4}$

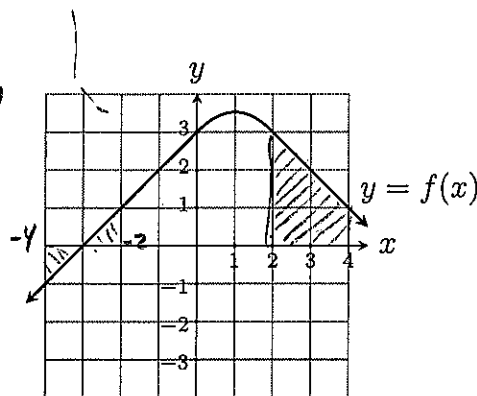
(b) $\int_{-4}^{-2} f(x) dx = A_{up} - A_{down} = \frac{1}{2} - \frac{1}{2} = \boxed{0}$

(c) $\int_3^4 f(x) dx = \boxed{0}$

(d) Suppose $\int_0^2 f(x) dx = 6.6$. Find $\int_0^4 f(x) dx$.

$$\int_0^4 f(x) dx = \int_0^2 f(x) dx + \int_2^4 f(x) dx = 6.6 + 4 = \boxed{10.6}$$

(e) $\lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(-3 + \frac{k}{n}\right) \frac{1}{n} = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k) \Delta x = \int_{-3}^{-2} f(x) dx = \boxed{\frac{1}{2}}$



$$a = -3$$

$$b = -3 + \frac{1}{n} = -2$$

$$\Delta x = \frac{-2 - (-3)}{n} = \frac{1}{n} \quad x_k = -3 + k \cdot \frac{1}{n}$$

2. Suppose for functions f and g we have: $\int_1^4 f(x) dx = 1$, $\int_4^6 f(x) dx = 2$, $\int_1^6 g(x) dx = 3$.Find $\int_1^6 (f(x) + 2g(x)) dx$

$$\begin{aligned} &= \int_1^6 f(x) dx + 2 \int_1^6 g(x) dx = \int_1^4 f(x) dx + \int_4^6 f(x) dx + 2 \int_1^6 g(x) dx \\ &= 1 + 2 + 2 \cdot 3 = \boxed{9} \end{aligned}$$

1. Answer the questions about the function $f(x)$ graphed below.

(a) $\int_{-4}^{-2} f(x) dx = A_{up} = \frac{1}{2} \cdot 2 \cdot 2 = \boxed{2}$

(b) $\int_2^2 f(x) dx = \boxed{0}$

(c) $\int_{-3}^0 f(x) dx = A_{up} - A_{down} = \frac{1}{2} - 2 = \boxed{-\frac{3}{2}}$

(d) Suppose $\int_1^2 f(x) dx = -2.3$. Find $\int_1^4 f(x) dx$.

$$\int_1^4 f(x) dx = \int_1^2 f(x) dx + \int_2^4 f(x) dx = -2.3 - 2 = \boxed{-4.3}$$

(e) $\lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(-4 + \frac{2k}{n}\right) \frac{2}{n} = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k) \Delta x = \int_{-4}^{-2} f(x) dx = \boxed{2}$

$$\begin{aligned}
 a &= -4 \\
 b &= -4 + \frac{2n}{n} = -2 \\
 \Delta x &= \frac{-2 - (-4)}{n} = \frac{2}{n}, \quad x_k = -4 + k \frac{2}{n}
 \end{aligned}$$

2. Suppose for functions f and g we have: $\int_1^7 f(x) dx = 4$, $\int_7^9 f(x) dx = 5$, $\int_1^9 g(x) dx = 6$.Find $\int_1^9 (f(x) - 3g(x)) dx$

$$\begin{aligned}
 &= \int_1^9 f(x) dx - 3 \int_1^9 g(x) dx = \int_1^7 f(x) dx + \int_7^9 f(x) dx - 3 \int_1^9 g(x) dx \\
 &= 4 + 5 - 3 \cdot 6 \\
 &= 9 - 18 = \boxed{-9}
 \end{aligned}$$

